ESCUELA DE CONSTRUCCIÓN CIVIL
FACULTAD DE INGENIERÍA
Revista de la Construcción
Journal of Construction

Research Article

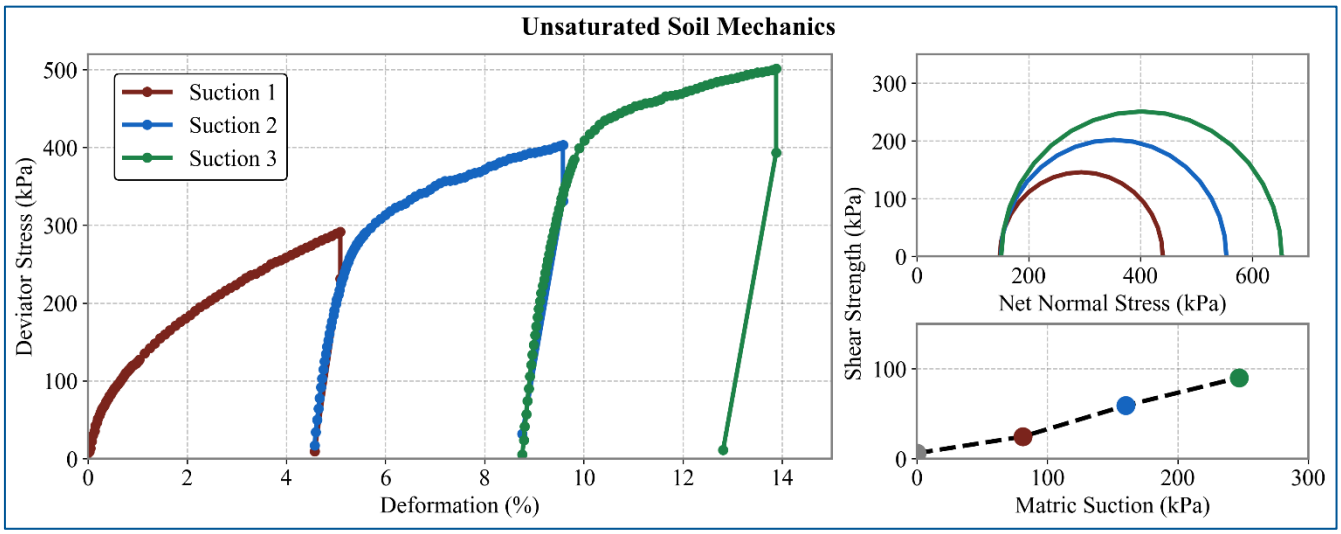
Influence of varying suction conditions on the shear strength of a clayey silt soil

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Received: 09.12.25; **Accepted:** 07.04.26; **Published:** 10.08.26

Citation: Lozano, W., Wilches, J., and Giacheti, H. (2026). Influence of varying suction conditions on the shear strength of a clayey silt soil. Revista de la Construcción. Journal of Construction, 25(2), 379-399. https://doi.org/10.7764/RDLC.25.2.379



Highlights:

- Unsaturated high plasticity silt studied via suction-controlled triaxial tests.
- Matric suction substantially increases shear strength and stiffness of the soil.
- Macrostructure and microstructure show high compressibility upon wetting.
- Reconstituted specimens underestimate in situ unsaturated soil behavior.
- PRISMA review supports conservative saturated strength parameter adoption.

Abstract: Unsaturated soil mechanics is a crucial area of geotechnical engineering that has gained increasing importance in recent years. However, despite its relevance, no standard guidelines have yet been implemented to study unsaturated behaviour in laboratories. This gap limits professionals' ability to address practical challenges in arid and tropical regions, particularly in problematic soils under shallow foundations, as well as in collapsible and expansive soils. In addition, critical geotechnical structures such as earth dams, slopes, and pavements require expertise in this field. In this context, the main objective of this study was to analyse the influence of varying suction conditions on the shear strength of a problematic clayey silt soil, which exhibits a macrostructure and microstructure with highly compressible characteristics and sensitivity to moisture variations. The research was conducted following ASTM standards for physical and compressibility characterisation. Specific methods included the consolidated drained triaxial compression test with controlled suction (CID-CA-CS) and the unconsolidated undrained (UU) triaxial test for unsaturated mechanical characterisation. The PRISMA model was also applied to define the strength parameters of saturated soil and estimate the linear failure envelope of unsaturated soil. The key findings include an increase in shear strength to 102.8, 128.8, and 157.4 kPa for matric suction values of 81.3, 160.2, and 246.9 kPa, respectively. This research concludes that suction plays a fundamental role in soil strength and stiffness. Additionally, when suction is lost, the soil undergoes significant deformations under loading. Finally, the reconstituted state of the material and unconsolidated undrained conditions are not representative to evaluate its unsaturated soil behaviour.

Keywords: Fine grained soils, matric suction, shear strength, unsaturated soils, macro/microstructure.

Abbreviation:

CID-CA-CS: Consolidated drained triaxial compression test with controlled suction
PRISMA: Preferred Reporting Items for Systematic Reviews and Meta-Analyses
UU: Unconsolidated-Undrained Triaxial Compression Test
c': Effective cohesion intercept, kPa
c: Cohesion intercept under unsaturated conditions, kPa
C_c: Compression ratio; also, coefficient of curvature
C_r: Recompression ratio (dimensionless)
C_u: Coefficient of uniformity (dimensionless)
$D\ (mm)$: Specimen diameter, mm
$D_{10}D_{30}D_{60}D_{85}$: Grain diameters (mm) at 10%, 30%, 60%, 85% passing
e: Void ratio (dimensionless)
$H\ (mm)$: Specimen height, mm
LL, PI, PL: Liquid limit (%), Plastic limit (%), Plasticity index (%)
MH: High plasticity silt (USCS symbol per ASTM D2487)
NR: Not reported
p^{0*}: Yield stress / preconsolidation pressure (kPa)
PSD: Particle size distribution
R^2: Coefficient of determination of linear regression
S_r, w: Degree of saturation (%), Gravimetric water content (%)
t_{90}: Time to 90% primary consolidation (min)
$USCS$: Unified Soil Classification System (ASTM D2487)
u_w, u_a: Pore water pressure (kPa), Pore air pressure, kPa
$(u_a - u_w)$: Matric suction, kPa
ϕ', ϕ^b: Internal friction angle (°), Suction dependent friction angle (°)
κ: Slope of unloading-reloading line in v–ln p space
λ: Slope of normal consolidation line in v–ln p space
σ_1, σ_3: Major principal stress (kPa), Minor principal stress (kPa).
τ: Shear stress, kPa

1. Introduction

Saturated soils play an important role in defining the geotechnical parameters in the earthworks projects in Peru, as well as in the soil mechanical characterization in the laboratory by conventional triaxial tests. However, various regions of the country exhibit a wide range of soils that require the practical application of unsaturated soil mechanics, including arid regions and tropical soils (Gil and Carrillo, 2015; Gil, 2016), collapsible and expansive soils (Rojas and Alva, 1988; Alva et al., 2020), as well as key infrastructures such as earth dams, shallow foundations on problematic soils, slopes (Torres Rodríguez, 2019) and pavements (Caicedo, 2019).

There are no standard guidelines currently in Peru for the mechanical characterization of partially saturated soils and there are no accredited laboratories on the subject (Alcances Acreditados por Laboratorios de Ensayo, Instituto Nacional de Calidad – INACAL, 2025), a fact that exacerbates the scarcity of research at this level of geotechnical engineering in the country. This lack significantly limits the advancement of knowledge in this field and conditions current research to use the principles of Terzaghi's effective stresses, designed for dry or saturated soils, which results in an oversimplification of the behaviour of partially saturated soils. Camapum et al. (2015) highlight the importance of unsaturated soils in geotechnical engineering, pointing out that, in land masses in tropical regions, with arid and semiarid climates, the unsaturated state is significant, which makes inappropriate the use of the initial concepts and theories of classical soil mechanics, which were elaborated for saturated or dry environments.

Unsaturated soils present several engineering challenges that distinguish them from their saturated or dry counterparts. Among the most critical are: (i) collapse upon wetting, wherein a metastable open fabric collapses under the combined action of overburden stress and suction loss (Fredlund and Rahardjo, 1993); (ii) swelling and heave, resulting from the expansion of active clay minerals upon moisture ingress, which can induce significant structural damage (Alonso et al., 1999); (iii) reduction in shear strength with increasing degree of saturation, as matric suction — which provides an apparent cohesion — decreases; and (iv) increased compressibility under loading after wetting, leading to differential settlements. These phenomena are particularly relevant in the problematic soils of northern Peru, where cyclical wetting, tropical climate and drying govern the geotechnical response of the subsoil.

For instance, the arid soils of the Peruvian coast are unsaturated due to the deep-water table, as the annual evaporation of the surface strata exceeds annual precipitation. According to Huerta and Lavado (2021), 11.5% of Peru's territory remains dry all year around, with semi-arid, arid, and hyper-arid zones experiencing prolonged periods of drought, sometimes all year round. These soils should therefore be studied within the framework of unsaturated soil mechanics.

Recent studies have explored this topic. Dong et al. (2024) investigated a cement-treated granite (GRS) exhibiting bimodal unsaturated hydraulic behaviour for subgrade and embankment applications. The authors found that cohesion and maximum shear strength of unsaturated cement-treated GRS samples increased with suction under constant net normal stress by using multi-stage triaxial shear tests. Similarly, Yang and Vanapalli (2024) conducted a series of suction-controlled ring shear tests on fine- and coarse-grained unsaturated soil samples subjected to large shear deformations. Their results demonstrated the influence of suction on residual strength, showing that apparent residual cohesion increases linearly with matric suction at a constant rate. In addition, they established a linear correlation between residual shear strength and the effective stress associated with matric suction and the effective degree of saturation. Furthermore, the effective residual friction angle obtained from all unsaturated samples was greater than that of the saturated samples.

Cai et al. (2022) investigated the critical state behavior of an unsaturated silty sand under constant matric suctions. Their findings indicate shear shrinkage at low dry soil densities and dilation at high soil densities. Regarding shear strength, it was positively correlated with dry density, matric suction, and net confining pressure. Similarly, Toll and Ali Rahman (2016) conducted triaxial tests on artificially unsaturated cemented sand to assess the applicability of various unsaturated soil frameworks for interpreting shear strength. They found that suction contributes to the strength of the cementing material itself, even in the critical state, and that changes in overall shear strength result from the desaturation of microvoids within the cementing material. Several international studies have advanced the understanding of unsaturated soil behaviour. Fredlund et

al. (1978) established the theoretical basis for the extended Mohr–Coulomb criterion, demonstrating that shear strength varies linearly with matric suction at a rate defined by the friction angle ϕ^b. Abd et al. (2020) demonstrated, using a modified triaxial device, that the relationship between shear strength and matric suction is bilinear, with an inflection point at the peak stress, and that matric suction also increases the Young's modulus of the soil. Song et al. (2024) proposed a shear strength equation valid over a wide suction range that accounts for the combined effects of capillarity and adsorption at various initial void ratios, demonstrating good predictive performance for plastic soils. Vanapalli et al. (1996) proposed a predictive model linking the soil-water characteristic curve to unsaturated shear strength, widely validated for fine-grained soils. More recently, Yang and Vanapalli (2024) established a linear correlation between residual shear strength and effective stress associated with matric suction and the effective degree of saturation from suction-controlled ring shear tests.

The theory proposed by Fredlund et al. (1978) is employed in this study to estimate the extended Mohr-Coulomb failure envelope for saturated soils incorporating the suction variable. That is, the estimated unsaturated strength envelope will be planar. In this regard, Ho and Fredlund (1982) concluded that the shear strength increases linearly with matric suction, a trend also observed by Rahardjo et al. (1994) in residual Singapore clay. However, it is important to note that Gan et al. (1988), Fredlund et al. (1995), Escario and Saez (1986), and Oloo and Fredlund (1996) have reported results indicating the non-linearity of the failure envelope in the shear stress versus matric suction plane. In this regard, the relationship between shear strength and matric suction has been shown to exhibit both linear and non-linear segments depending on the suction magnitude, with suction also increasing the elastic stiffness of the soil (Abd et al., 2020). More recently, Song et al. (2024) proposed a shear strength equation valid over a wide suction range that accounts for the joint contribution of capillarity and adsorption at various initial void ratios, demonstrating improved predictive performance over classic linear models for plastic soils.

An analysis of the influence of varying suction conditions on the shear strength of a problematic clayey silt soil will be conducted, and the unsaturated failure surface will subsequently be estimated. The novelty of the present study lies in the characterisation of the macrostructural and microstructural compressibility of a high plasticity silt soil (MH) from the Cajamarca region of Peru, a soil type that has received limited attention in the international literature, on undisturbed specimens, preserving the in-situ fabric and structural arrangement that govern the unsaturated mechanical response. This is achieved using a multi-stage consolidated drained triaxial test with controlled suction (CID-CA-CS) and the axis translation technique. Furthermore, the study integrates the PRISMA systematic review protocol, which provides a structured and reproducible data compilation that supports a more rigorous and objective analysis of strength parameters, offering a replicable methodological framework for research on unsaturated fine-grained soils in data-scarce regions.

In line with this novelty, the study pursues four specific objectives. The first is to characterize the physical, compressibility, and shear strength behaviour of an undisturbed high plasticity silt (MH) from the Cajamarca region under saturated and suction-controlled conditions. The second is to evaluate, through multi-stage CID-CA-CS triaxial tests with the axis translation technique, the influence of matric suction on shear strength and stiffness, and to estimate the corresponding extended Mohr–Coulomb failure envelope. The third is to contrast the mechanical response of undisturbed specimens, which preserve the in-situ fabric, against reconstituted specimens and UU test results, assessing how representative each approach is for unsaturated soil characterisation. The fourth is to apply the PRISMA systematic review protocol to compile saturated strength parameters from the literature, supporting a conservative and reproducible estimation of the failure surface. The paper is organized accordingly: Materials and methods details the studied soil, sampling procedure, and testing programme, including the systematic review protocol; Results and discussion presents the soil's behaviour under saturated and unsaturated conditions and discusses the effects of suction loss and sample disturbance; and Comments and conclusions summarizes the main findings, states the study's limitations, and outlines future research directions.

2. Materials and methods

2.1. *Materials*

The studied soils are located in the province of Cutervo (Latitude: 6°10'17.90" S, Longitude: 78°43'1.02" O) in the department of Cajamarca, in northern Peru. The samples were collected at a depth of 1.50 meters and comprised both undisturbed cubic blocks and disturbed samples for physical characterisation. Physical characterization adhered to standardized methodologies to ensure the reliability of the test data. Sample preservation and transport were conducted in accordance with ASTM D4220, which outlines procedures to maintain the integrity of soil specimens from field collection to laboratory analysis. The specific gravity of soil solids was determined following ASTM D854 (Procedure B), employing the water displacement method with a pycnometer or volumetric flask (minimum capacity: 250 mL). Particle size distribution (PSD) was determined using ASTM D6913 for coarse soil fractions (mechanical sieving) and ASTM D7928 for fine soil fractions (sedimentation analysis via hydrometer, based on Stokes' law). From the particle size distribution curves, the characteristic diameters were determined: D_{10}, D_{30}, D_{60}, and D_{85} represent the grain diameters (in mm) at which 10%, 30%, 60%, and 85% of the soil mass, respectively, is finer. These parameters are used to compute the coefficient of uniformity ($C_u = D_{60}/D_{10}$) and the coefficient of curvature ($C_c = {D_{30}}^2 / (D_{10} \times D_{60})$), required for USCS classification under ASTM D2487. Additionally, ASTM D2216 was employed to determine the gravimetric water content of the soil at 110 °C, while ASTM D4318 was utilized to determine the Atterberg limits, including the liquid limit, plastic limit, and plasticity index. Soil classification for engineering purposes was conducted in accordance with the Unified Soil Classification System (USCS) as per ASTM D2487. ASTM D7263 was applied to quantify the bulk density and unit weight of the analysed sample. Table 1 summarizes the soil physical properties.

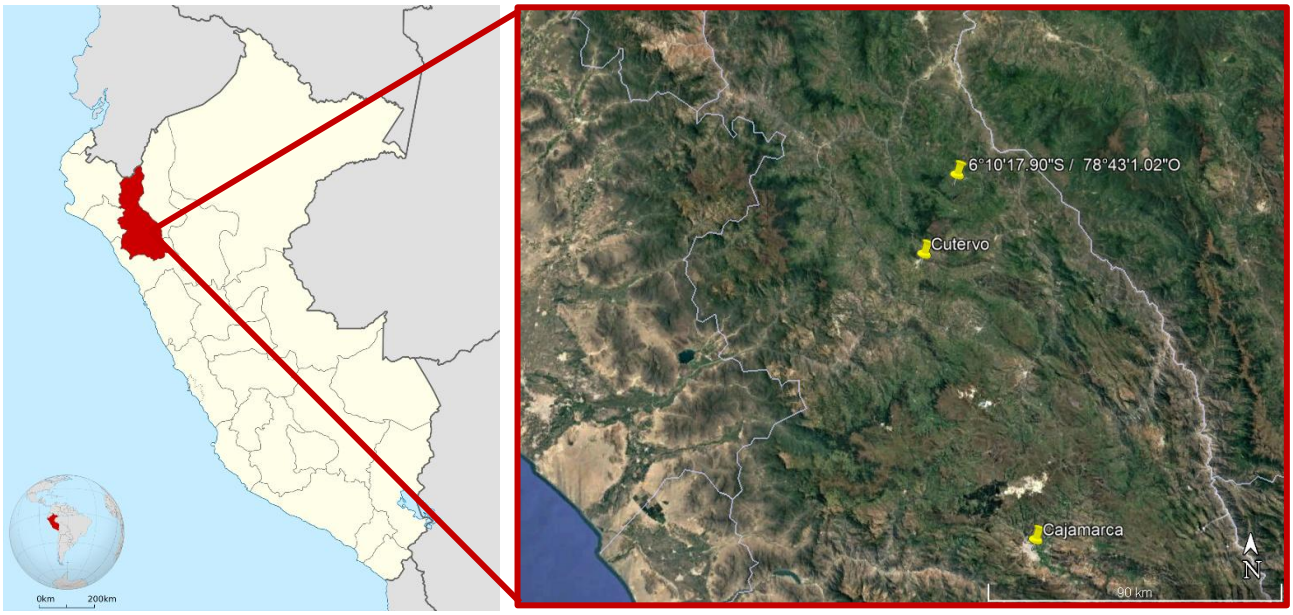


Figure 1. Geographical location and sampling site in the province of Cutervo (6°10'17.90" S, 78°43'01.02" W).

Table 1. Physical characterization of the soil.

USCS	Sand (%)	Silt (%)	Clay (%)	Specific mass of solids (Mg/m^3)	Dry specific mass (Mg/m^3)	Natural moisture content (%)	Liquid limit (%)	Plasticity index (%)
MH	1	58	41	2.70	0.987	55	75	30

It is important to note that during the sample extraction, clasts with very compressible characteristics were observed, which makes this soil problematic, since both the macrostructure and microstructure play a fundamental role in the saturated and

unsaturated behavior when loads are applied. On the other hand, these unique features do not allow generalization of this study. In this respect, Alonso et al. (1999) mention that when studying a cohesive soil with expansive characteristics and collapsible structure, it is extremely difficult to generalise the results obtained at a specific site with other soils of similar classification. According to Gens and Alonso (1992), this type of soil has two distinct levels of structure: the microstructural level, where the expansion of active minerals takes place, and the macrostructural level, responsible for the main structural reorganisation. This aspect can be observed in Figure 2. It can also be seen in this figure the longitudinal and transverse compressible macrostructure of the soil sample.

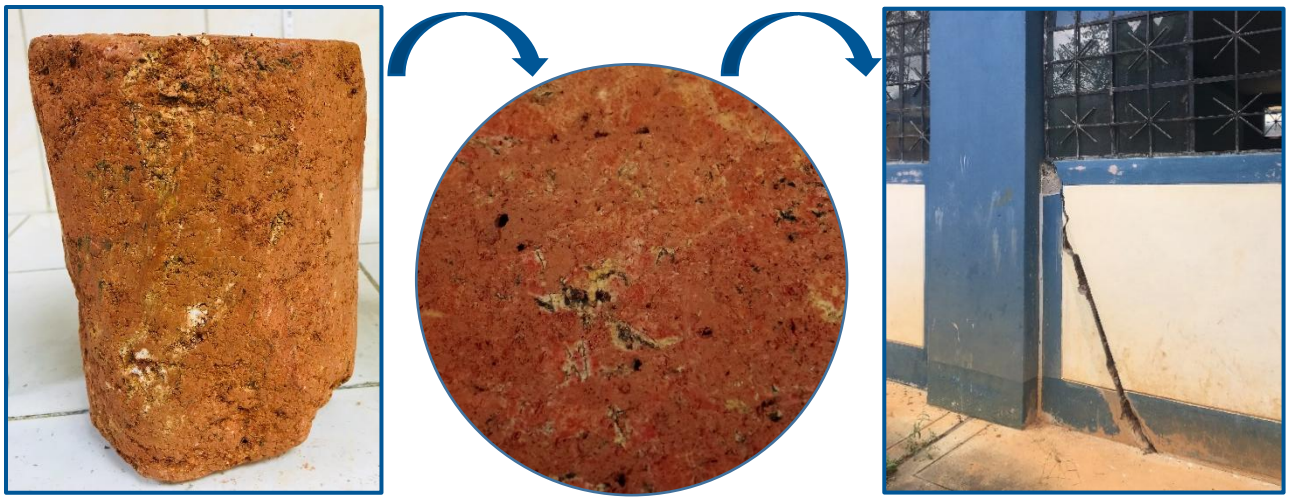


Figure 2. Field sampling specimen showing the loss of soil suction and associated settlement and collapse problem due to the presence of a plastic microstructure and highly compressible macrostructure.

The soil's particle size distribution in Figure 3 shows that nearly 100% is fine (passes the 0.075 mm sieve), with a plasticity index of 45% (ASTM D4318, 2017).

2.2. *Methods*

A series of advanced geotechnical tests were conducted to characterize the mechanical behavior of the soil under varying saturation and loading conditions. A consolidated drained triaxial compression test with controlled suction (CID-CA-CS) was performed, adapted from established methodologies in the scientific literature and widely employed in research laboratories for the mechanical characterization of unsaturated soils.

Furthermore, the assessment of macrostructural and microstructural behaviour needs one-dimensional consolidation and collapse potential tests to be carried out in accordance with ASTM D4546 and ASTM D2435 to evaluate the soil's deformation response to incremental loading under saturated conditions. In addition, an unconsolidated undrained triaxial compression test (UU) was conducted under partially saturated conditions, following an adapted methodology based on ASTM D2850, to examine the variation in shear strength under constant suction and changing normal stress conditions. Suction measurements were obtained by using the filter paper method as per ASTM D5298.

As the CID-CA-CS test is not standardized in Peru, a step-by-step experimental procedure was developed based on the methodology outlined by Julca Lozano (2025). This method integrates the ASTM D7181-20 standard for consolidated drained triaxial compression tests with the axis translation technique introduced by Hilf (1956). The detailed execution process is presented in the following section.

2.2.1. Specimen preparation

The specimen was extracted from an intact cubic block using a sample extruder, yielding a cylindrical sample with a diameter of 71 mm and a height of 142 mm (Figure 4), ensuring a height-to-diameter ratio between 2.0 and 2.5. The maximum particle size was maintained below one-sixth of the specimen diameter, in accordance with standard testing protocols. The height-to-diameter ratio was selected to comply with ASTM D7181-20 and to ensure a well-defined failure plane with minimal end-restraint effects (Lade, 2016). The specimen diameter of 71 mm was chosen such that the maximum particle size does not govern specimen dimensions (max. particle size ≤ D/6), consistent with the requirements of ASTM D7181-20 and with the specimen dimensions used by Banerjee et al. (2020) for similar fine-grained unsaturated soils.

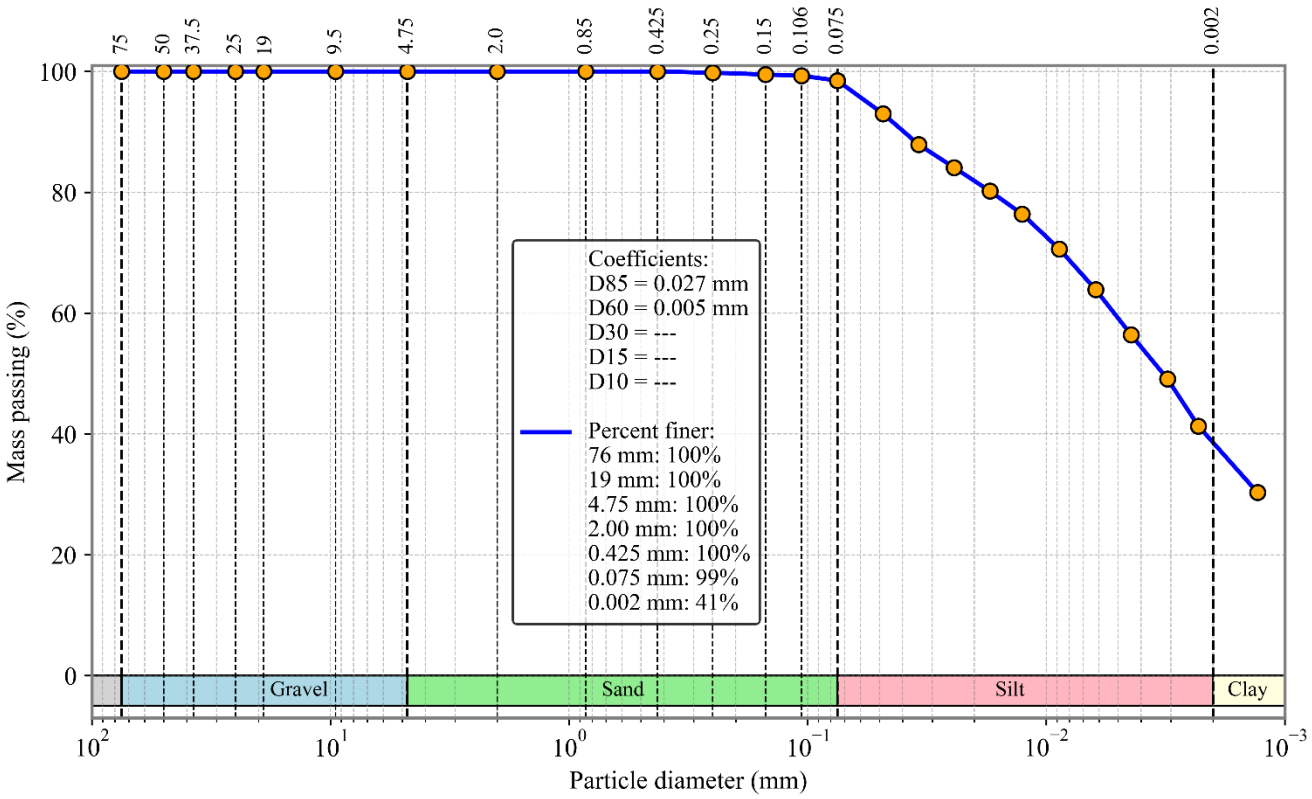


Figure 3. Summary of grain size distribution analysis | ASTM D6913−17 and ASTM D7928−21.



Figure 4. Test specimen preparation.

2.2.2. Preliminary saturation phase

Prior to specimen preparation, both the silicon carbide porous stone and the high air-entry ceramic disc were saturated with deionized water for at least 24 hours. A filter paper disc was placed on both the upper and lower bases of the specimen, with the lower disc resting on the base pedestal of the triaxial cell; this step served to reduce the matric suction of the specimen of in-situ field conditions. The specimen was then enclosed within a latex membrane to prevent lateral expansion during the initial saturation phase, and the membrane was secured with O-rings at the top and bottom contacts with the triaxial cell. To enhance saturation efficiency, the specimen was de-aired and subsequently submerged in a deionized water-filled chamber for 48 hours.

2.2.3. Back pressure saturation procedure

The aim of this stage was to fill all remaining voids in the specimen with deaerated water under controlled conditions. The upper platen was first secured with O-rings, after which the triaxial cell was assembled and the confining chamber was filled with water, ensuring that no uncontrolled stresses were induced. The drainage system was then fully saturated by regulating the flow through the pressure control panel's valves and burettes. Incremental confining pressures were applied to both the cell and the specimen to facilitate progressive saturation. One of the specimen's drainage outlets was connected to a pressure transducer or manometer to monitor the initial pore water pressure. Upon stabilization of the pore pressure or burette readings, the backpressure saturation process was initiated. Skempton's B-parameter (Skempton, 1954) was determined at each increment to evaluate the degree of saturation, ensuring compliance with standardized criteria before proceeding to the next phase.

2.2.4. Stress application

The test is of the multi-stage type. The series of stresses defined for Stages 1, 2 and 3 are shown in Table 2. In this process, the aim is to apply constant net normal stress and varying matric suction levels to evaluate their effect on shear strength.

Table 2. Set of applied stresses for CID-CA-CS test.

Indicators	Stage 1 (kPa)	Stage 2 (kPa)	Stage 3 (kPa)
σ_3	275	362.5	450
u_w	50	50	50
u_a	125	212.5	300
$(\sigma_3 - u_a)$	150	150	150
$(u_a - u_w)$	75	162.5	250
$H_0\ (mm)$	155.12	147.21	139.70
$D_0\ (mm)$	72.71	69.00	65.48

2.2.5. Hydrostatic consolidation

This step allowed equilibrium to be reached under drained conditions, matching the stresses applied in Stages 1, 2, and 3. Matric suction was maintained constant throughout both the hydrostatic consolidation and the drained triaxial shear stages by means of the axis translation technique, using independent air pressure (u_a) and back-pressure (u_w) controllers. The target suction levels ($u_a - u_w$ = 75, 162.5, and 250 kPa) were applied at the beginning of each stage and monitored continuously until equilibrium before shearing. Additionally, data were obtained to determine the shear rate. The shear rate was determined from the time to 90% primary consolidation (t_{90}) following ASTM D7181-20. The selected rate of 0.0017 mm/min corresponds to an axial strain rate of approximately 0.0011 %/min. This is within the range recommended for drained testing of fine-grained soils (Ho and Fredlund, 1982). Upon application of Stage 1 stresses, appropriate drain valves were opened to allow the specimen to drain into the volume change burette. Volume changes were monitored and recorded at time intervals of 0.1, 0.2, 0.5, 1, 2, 4, 8, 15 and 30 minutes and at 1, 2, 4, 8, 16, 24 and 48 hours. Volume change and strain gauge readings were then plotted against the logarithm or square root of the elapsed time to determine the time to 90% primary consolidation (t_{90}). The degree of saturation (S_r) of the specimen was estimated at each stage from the volumetric water content and total void ratio, calculated from the initial specimen dimensions, the gravimetric water content at specimen preparation, and the volume changes recorded during consolidation. S_r was not directly measured during shearing; however, the suction values applied via the axis translation technique implicitly constrain the air–water interface state, and the near-constant volume changes observed during shearing indicate that S_r remained approximately stable throughout the drained shear phase.

2.2.6. Drained triaxial shearing with controlled suction

For the suction-controlled (constant suction) triaxial shear stage, GEOTAC's advanced automated TruePath system was employed. This system integrates state-of-the-art data acquisition and feedback control mechanisms for the test. The TruePath system comprises a loading frame, two DigiFlow pressure-volume actuators (flow pumps), and a high-resolution analogue data acquisition system, all interfaced with a computer through the TestNet data acquisition platform. Additionally, the GEOTAC Flex Panel was utilized to apply air pressure (Figure 5).

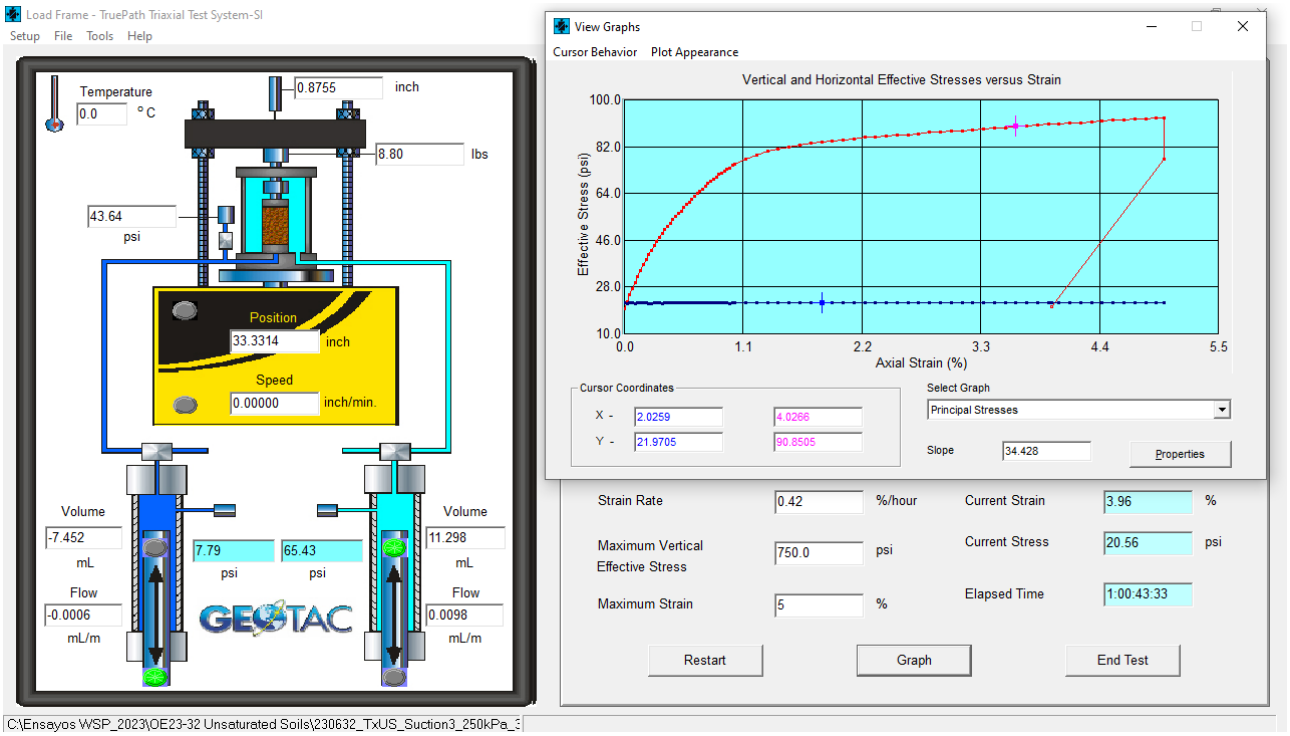


Figure 5. Software interface - triaxial CID-CA-CS - Stage 3. Source: WSP laboratory in Peru.

2.2.7. Calculations

The calculations in this experimental procedure were performed under the following formulations:

$$\text{Net normal stress} = (\sigma - u_a) \quad (1)$$

where σ = total normal stress on the specimen (kPa); u_a = pore air pressure (kPa); $(\sigma - u_a)$ = net normal stress, the stress state variable associated with the soil skeleton in unsaturated conditions (Fredlund and Morgenstern, 1977).

$$\text{Shear stress} = \left(\frac{\sigma_1 - \sigma_3}{2}\right) \quad (2)$$

where σ_1 = major principal total stress at failure (kPa); σ_3 = minor principal total stress (kPa); $(\sigma_1 - \sigma_3)/2$ = deviator net stress expressed as shear stress on the p–q plane.

$$\text{Mean net stress} = \left(\frac{\sigma_1 + \sigma_3}{2}\right) - u_a \quad (3)$$

where σ_1 = major principal total stress at failure (kPa); σ_3 = minor principal total stress (kPa); u_a = pore air pressure (kPa); and $((\sigma_1 + \sigma_3) / 2) - u_a$ = mean net stress on the p–q plane, representing the average of the major and minor principal net stresses acting on the specimen under the axis translation framework (Fredlund and Morgenstern, 1977).

$$\text{Matrix suction} = (u_a - u_w) \quad (4)$$

where u_a = pore air pressure (kPa); u_w = pore water pressure (kPa); $(u_a - u_w)$ = matric suction, the component of total suction due to capillary and adsorptive forces at the air–water interface (Fredlund and Rahardjo, 1993).

2.3. *Systematic review*

To design the failure surface of unsaturated soils, at least two series of triaxial tests were necessary: one in saturated and one in unsaturated conditions. In this study, two unsaturated series of the drained and the unconsolidated undrained type. For this reason, a systematic search for records containing typical shear strength parameters of the type of the study soil was carried out and a conservative value was adopted to estimate the failure surface.

The systematic search was conducted using the Springer Link and ScienceDirect databases, with records restricted to publication from 2020 to the present. The Boolean search query ((Triaxial Test for Unsaturated Soils) AND (Suction)) retrieved 148 records from Springer Link and 169 from ScienceDirect. After title screening, 304 records were excluded, and one duplicate entry was removed.

There were three exclusion criteria for the records. The number of records removed is shown in brackets.

- Criterion 01: For not containing triaxial tests applied to unsaturated soils (n = 7).
- Criterion 02: For not dealing with fine soils (n = 2).
- Criterion 03: For being out of context and not relating strength and suction (n = 1).

On the other hand, the manual search consisted of selecting papers from the book The Emergence of Unsaturated Soil Mechanics where the mechanics of unsaturated soils at depth is studied. Two records were included, both written by Ho and Fredlund (1999). The selected records are: A multistage triaxial test for unsaturated soils and increase in strength due to suction for two Hong Kong soils.

Exceptionally, the work of Fernandes (2022) was also included in the manual search, given that the author employed a similar suction-controlled triaxial test methodology on undisturbed specimens and provided an extensive dataset of unsaturated shear strength parameters that are directly comparable to those of the study soil. Notably, the results consistently show an increase in the cohesion intercept with matric suction across all depths and conditions, while no significant variation in the internal friction angle was observed. The strength parameters reported by this author are presented in Table 3. Figure 6 shows the flow chart for the search of records using the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) method.

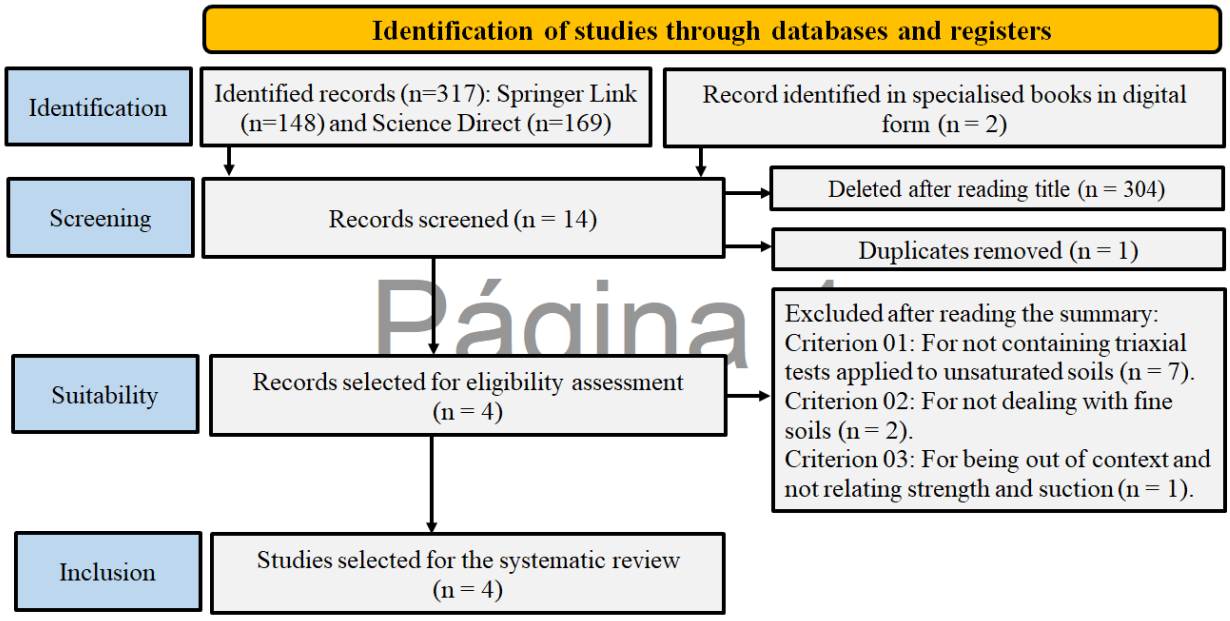


Figure 6. Flow chart for the systematic review applying the PRISMA model.

3. Results and discussion

Method B of the aforementioned standard was employed to measure the one-dimensional collapse or swelling potential on an intact cohesive soil specimen, induced by wetting or hydrocompression at a normal confining stress of 20 kPa. The unsaturated soil exhibited minimal compressive deformation values (Figure 7) up to this stress, due to the macrostructure. However, the microstructure was activated upon soil inundation producing expansion deformations. The measured swelling potential at this stress level was 0.14%, indicating proximity to the swelling pressure.

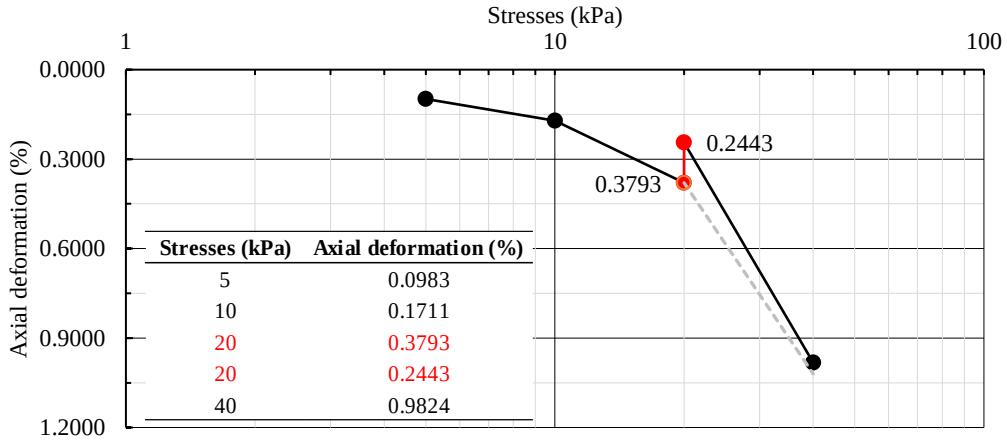


Figure 7. One-dimensional collapse with swelling or collapse potential.

The one-dimensional consolidation test encompassed a range of applied axial stresses from 5 kPa to 1280 kPa under saturated conditions, in accordance with the loading sequence specified by ASTM D2435 (2020). The objective was to activate both the macrostructure and microstructure to evaluate their response to loading. The soil exhibited significant deformations below the swelling pressure, reaching a total deformation exceeding 20% under a vertical stress of 1280 kPa. Soil parameters obtained from the test include yield stress (preconsolidation pressure, p_0^*), compression ratio (c_c) and recompression ratio (c_r), as shown in Figure 8.Additionally, the constitutive parameters lambda (λ) and kappa (κ) were determined to facilitate the understanding and prediction of soil behavior in the critical state and during loading and unloading cycles, for both saturated and unsaturated conditions, when tested under controlled suction. According to Hazelton and Murphy (2016), the soil's microstructure exhibits high plasticity, which corresponds to a high degree of compressibility.

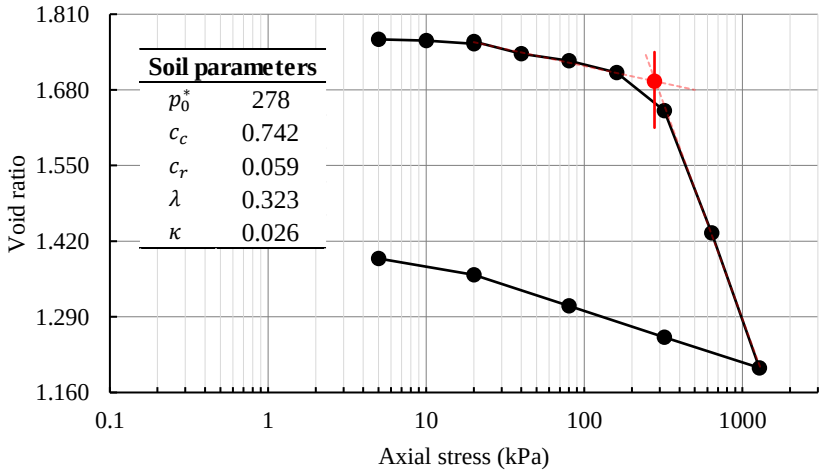


Figure 8. One-dimensional consolidation test.

Based on the data of the systematic review, a friction angle of 25° under saturated conditions is adopted for in this study along with an effective cohesion intercept of 6 kPa due to the higher clay content compared to the findings of Zheng et al. (2020) and Banerjee et al. (2019). It becomes possible to estimate all strength parameters of the unsaturated soil based on these assumptions, including the suction-dependent friction angle, the variation of the cohesion intercept for each change in applied matric suction, and the linear 3D failure surface. Table 3 presents the parameters of various soils that follow a similar approach, exhibiting a linear variation of the cohesion intercept with respect to matric suction.

Table 3. Authors, soils, characteristics and parameters from the literature review.

Autor	Soil type	$u_a - u_w$	c'	ϕ'	c	ϕ^b
Zheng et al. (2020)	Clay fraction = 21 % Silty fraction = 78 % Sand fraction = 1 % Liquid limit = 32 % Plasticity index = 11 %	0	4.1	25.5	-	-
		100	4.1	25.7	51.7	25.5
		200	4.1	26.3	93.4	24.2
		300	4.1	26.0	136.7	23.9
Banerjee et al. (2019)	Clay fraction = 10 % Silty fraction = 86 % Sand fraction = 4 % Liquid limit = 25 % Plasticity index = NP	0	2.7	33.6	-	-
		50	2.7	33.5	30.8	29.3
		250	2.7	76.3	32.7	16.4
		0	3.5	33.4	-	-
		50	3.5	32.6	33.2	30.2
		250	3.5	76.8	32.1	16.3
Ho and Fredlund (1999a)	Decomposed granite	0	0.0	33.4	-	-
		241	0.0	33.4	90.0	12.4
		483	0.0	33.4	135.0	12.4
		0	32.5	33.4	-	-
		241	32.5	33.4	50.0	22.5
		483	32.5	33.4	76.0	22.5
	Decomposed rhyolite	35	-	35.3	70.0	10.9
		69	-	35.3	96.0	10.9
		138	-	35.3	110.0	10.9
		35	-	35.3	45.0	15.9
		69	-	35.3	77.0	15.9
		138	-	35.3	104.0	15.9
Ho and Fredlund (1999b)	Brownish pink, dense, slightly clayey, sandy silt with some fine gravel. (Completely weathered granite)	NR	-	33.4	-	12.4
	Pink with grey spots, dense to very dense, silt and sand with some fine gravel. (Completely to highly weathered granite)	NR	-	-	-	27.4
	Pink with grey spots, dense, sandy silt with some fine gravel. (Completely weathered granite)	NR	-	-	-	14.6
	Brownish pink, medium dense, slightly clayey, sandy silt with some fine gravel. (Completely weathered granite)	NR	-	33.4	-	16.2
	Brownish, medium to fine, slightly clayey, sandy silt. (Decomposed volcanic rock)	NR	-	35.3	-	13.5
	Light yellowish grey with some white, dense (slightly clayey), silt and fine sand (with some medium and coarse sand). (Completely weathered volcanic rock)	NR	-	35.3	-	13.3

Autor	Soil type	$u_a - u_w$	c'	ϕ'	c	ϕ^b
	Light yellowish grey with some white, dense (slightly clayey), silt and fine sand (with some medium and coarse sand). (Completely weathered volcanic rock)	NR	-	35.3	-	11.7
	Yellowish brown, dense silty sand with some fine gravel. (Completely weathered granite)	NR	-	33.4	-	11.5
	Orange brown, medium dense sandy silt. (Completely weathered granite)	NR	-	-	-	4.2
	Reddish brown, silty sand with patches of white, plastic, fine clay. (Completely weathered granite)	NR	-	33.4	-	12.5
	Reddish brown, silty sand with patches of white, plastic, fine clay. (Granitic residual soil)	NR	-	33.4	-	12.6
	Reddish brown with dark green and white specks, sandy silt with some fine gravel and patches of plastic clay. (Granite residual soil)	NR	-	33.4	-	22.5
	Yellowish brown with dark green specks, dense sandy silt. (Completely weathered granite)	NR	-	-	-	8.0
Fernandes (2022)	USCS Classification: SC Tropical classification: Sandy Laterite (LA') Depth: 1.5 m	50, 200, 400	0.0	29.9, 27.8, 24.6	3, 11, 16	2.5
	USCS Classification: SC Depth: 3.0 m	50, 200, 400	1.2	33.5, 33.8, 33.8	6.5, 13.4, 20.5	2.9
	USCS Classification: SC Depth: 5.0 m	50, 200, 400	5.3	33.7, 34.0, 34.9	10.3, 24.2, 28.7	3.8
	USCS Classification: SC Depth: 7.0 m	50, 200, 400	5.0	31.1	23.5	7.7
	USCS Classification: SC Depth: 9.0 m	50, 200, 400	5.2	35.3, 34.6, 29.5	15.0, 36.0, 89.0	11.2
	USCS Classification: SC Depth: 11.0 m	50, 200, 400	9.5	37.6	31.0	12.3
	USCS Classification: SC Depth: 13.0 m	50, 200, 400	3.4	35.2, 35.6, 27.2	17.0, 48.0, 112.0	14.7
	USCS Classification: SC Depth: 16.0 m	50, 200, 400	2.0	32.8, 31.2, 28.3	31.0, 72.0, 123.0	17.5

Notes: NR = Not reported. Suction values were not reported.

The unsaturated soil parameters are shown in Figure 9. Three stages of matric suction were applied with constant measured values of 81.3 kPa, 160.2 kPa and 246.9 kPa. These values were controlled and monitored during the hydrostatic consolidation and triaxial shear phases at a single effective stress level of 150 kPa (net normal stress). An increase in stiffness was observed with rising suction based on the calculated elastic modulus from the stress-strain curves of the triaxial test.

A significant increase of 141% was noted in the cohesive intercept form the first to the second stages, followed by a moderate increase of 52% during the third stage. Shear stress, meanwhile, increased by 25% in the second stage and by 22% in the third stage. These findings align with the trend observed by Ho and Fredlund (1999). Under applied suctions of 50, 150, and 250 kPa, they reported a 37% increase in cohesion from the first to the second stage, followed by a further 15% increase during the third stage in decomposed granite. The same authors recorded cohesion increases in a decomposed rhyolite of 71% and 35% for the second and third stages, respectively.

Similarly, Banerjee et al. (2019) observed that increasing suction from 50 to 250 kPa enhanced cohesion by 136% for an ML soil (clayey silt). Zheng et al. (2020) reported that increasing suction for an ML soil type from 100 to 200 and 300 kPa led to an 81% rise in cohesion between the first and second stages, followed by a 46% increase in the third stage.

Once the parameters have been determined, the extension of the Mohr-Coulomb failure envelope for saturated soils to incorporate suction was applied for the study soil, as proposed by Fredlund et al. (1978). The equation is based on the two independent and uncoupled stress state variables $\sigma - u_a$ and $u_a - u_w$, proposed by Fredlund and Morgenstern (1977), as the most appropriate. Note the extended equation (5) for unsaturated soils and the estimated failure surface in Figure 9.

$$\text{Extended Mohr–Coulomb failure envelope: } \tau = c' + (\sigma - u_a) \cdot tan\phi' + (u_a - u_w) \cdot tan\phi^b \qquad (5)$$

where: τ = shear stress at failure (kPa); c' = effective cohesion intercept under saturated conditions (kPa); σ = total normal stress on the failure plane (kPa); u_a = pore air pressure (kPa); ϕ' = effective internal friction angle under saturated conditions (°); u_w = pore water pressure (kPa); ϕ^b = friction angle indicating the rate of increase in shear strength with matric suction (°). The term $(u_a - u_w) \cdot tan\,\phi^b$ represents the apparent cohesion contribution from matric suction.

Figure 10 presents the estimated three-dimensional failure surface for the study soil in the extended Mohr–Coulomb space defined by the net normal stress $(\sigma - u_a)$, the matric suction $(u_a - u_w)$, and the shear stress (τ) axes. The planar failure surface was constructed using the strength parameters adopted from the systematic review (c' = 6 kPa, ϕ' = 25°) and the suction friction angle ϕ^b back-calculated from the CID-CA-CS test results. The surface clearly illustrates that shear strength increases simultaneously with both net normal stress and matric suction, confirming the fundamental premise of the extended Mohr–Coulomb framework (Fredlund et al., 1978). The failure envelope is planar within the tested suction range (81.3–246.9 kPa), consistent with the linear relationship reported for similar high-plasticity silty soils by Ho and Fredlund (1999) and Banerjee et al. (2020), and further corroborated by the bilinear relationship documented by Abd et al. (2020) for the low-to-moderate suction range. Extrapolation beyond the tested range may result in non-linearity of the failure envelope, as documented by Gan et al. (1988), Escario and Saez (1986), and Song et al. (2024).

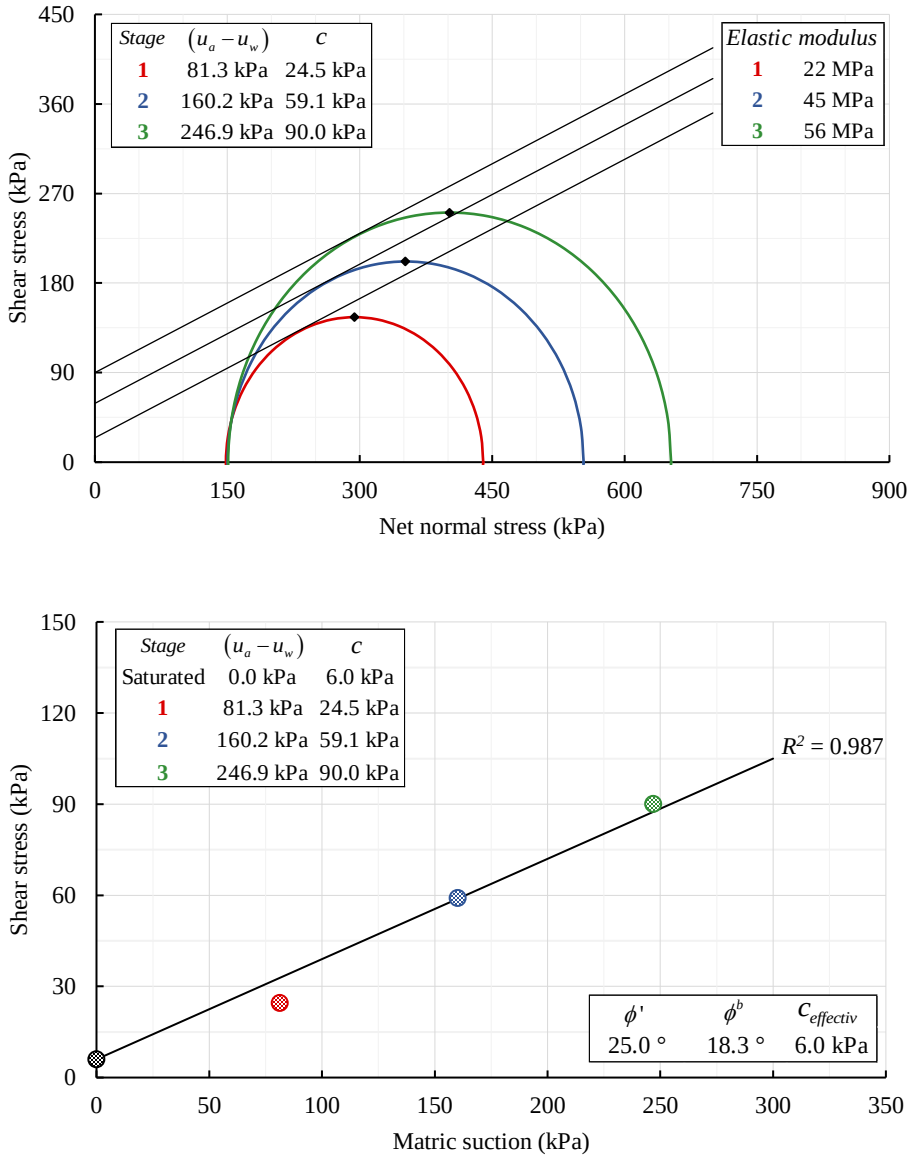


Figure 9. Unsaturated soil parameters for the investigated soil.

In response to the changes in strength of a partially saturated soil subjected to constant suction and variations in confining stress, three cylindrical specimens were remolded under in-situ conditions and tested using unconsolidated undrained (UU) triaxial compression, also referred to as rapid testing procedure. The resulting shear strengths for confined stresses of 100, 200, and 400 kPa were 41.7, 47.2, and 61.4 kPa, respectively (Figure 11). An average matric suction of 26 kPa resulted in 3.8°inclination angle. On the one hand, the variation in strength under different confining stresses confirms that the average effective stress governs the shear strength behavior of unsaturated soils, as established by Rasool and Kuwano (2018). On the other hand, the mechanical behavior demonstrates a distinct response since the variable suction was not controlled during these tests. Finally, a crucial aspect is the drastic change in soil structure caused by remoulding. This change is reflected in the strength values, when the stresses from these UU tests are applied to Equation 5, the strengths for the three confined stresses levels reach only 68%, 44%, and 31% of the estimated failure surface.

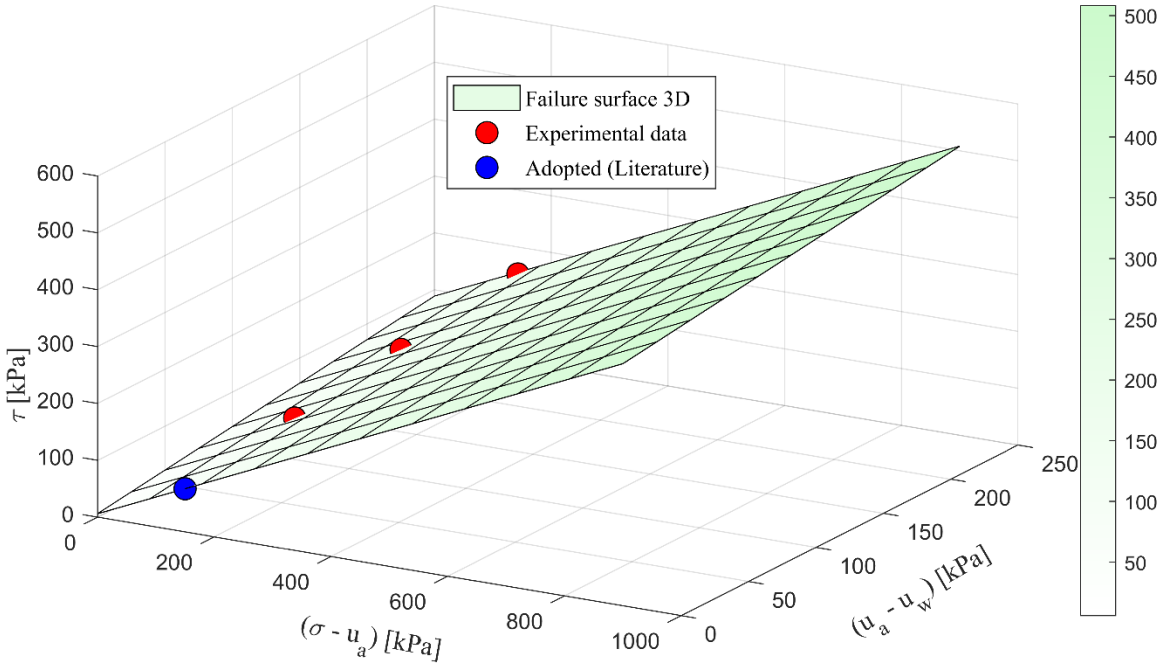


Figure 10. Estimated three-dimensional failure surface for unsaturated soils.

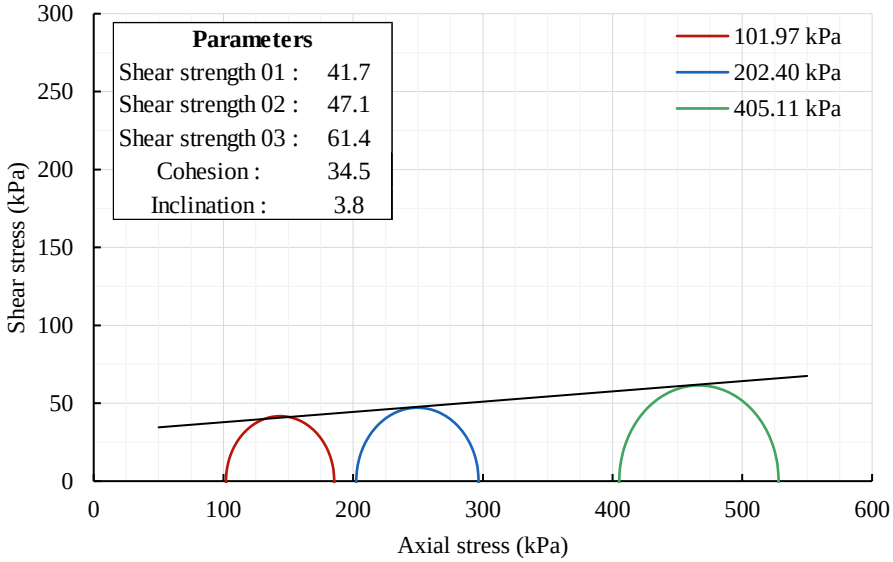


Figure 11. Influence of confining stress and change of soil structure.

The confined stress values reported in Figure 10 differ slightly from the target values of 100, 200, and 400 kPa because the GEOTAC pressure control system operates natively in psi units. The reported values result from the unit conversion to the International System of Units (SI) and the calibration corrections applied during data reduction. This procedure ensures greater precision in the reported stress state and does not affect the validity of the results.

4. Comments and conclusions

The study of the influence of varying soil suction conditions on the shear strength, unsaturated compressibility, and mechanical behavior leads to the following comments and conclusions:

1. The geotechnical characterization of the soil classified it as MH (high plasticity silt) according to the USCS system, with a predominance of fine particles, primarily silt and clay. This composition suggests a high susceptibility to moisture and/or suction changes, which significantly influence its geotechnical behavior.

2. The studied soil has a highly compressible macrostructure due to the presence of decomposing red shales in the subsurface, combined with a plastic microstructure characterized by a high degree of compressibility, as indicated by the plasticity index. Consequently, it tends to expand until reaching an equilibrium stress of approximately 20 kPa when the soil loses suction. Large deformations occur under higher loads, leading to differential settlement in structures and collapse upon wetting. Additionally, the study soil showed evidence of overconsolidation.

3. The systematic literature review enabled the conservative adoption of an effective cohesion value of 6 kPa and an internal friction angle of 25° for the saturated condition of the study soil. These parameters facilitated estimation of the failure surface using the extended Mohr–Coulomb equation for unsaturated soils. For experimental suction values of 81.3, 160.2, and 246.9 kPa, cohesion intercepts of 24.5, 59.1, and 90.0 kPa were obtained, respectively. The corresponding shear strengths were 102.8, 128.8, and 157.4 kPa. These findings emphasize the fundamental role of suction in enhancing soil stiffness and strength.

4. The unconsolidated undrained triaxial tests assessed the effect of suction on the soil's shear strength, demonstrating a notable increase in this property with higher effective confined stress values. However, this methodology is inadequate for accurately evaluating the behavior of partially saturated soils due to the lack of control over pore water distribution and the degree of saturation during the shear phase. Moreover, the reconstituted state of the sample significantly altered its structure, yielding results that do not accurately represent in-situ conditions.

These findings underscore the critical importance of considering moisture and suction conditions in the geotechnical analysis of partially saturated soils, particularly for the problematic soils beneath shallow foundations, as well as expansive and collapsible soils common in arid and tropical regions of the Earth. Such considerations are also essential for critical applications, including downstream slopes of earth dams, slope covers, embankments, and pavements.

This study is subject to some limitations that should be acknowledged. First, the results correspond to a single soil profile from one site in the Cajamarca region, so the findings and the estimated strength envelope cannot be directly extrapolated to other high plasticity silts without additional testing. Second, the CID-CA-CS test with controlled suction lacks a standardized procedure, which required adapting testing methods from the international literature. Third, the saturated strength parameters used to estimate the extended Mohr–Coulomb envelope were adopted conservatively from a systematic literature review rather than derived exclusively from tests on the study soil, which may introduce some uncertainty into the estimated failure surface. Future research should extend this characterisation to a broader range of high plasticity silts and stress paths, incorporate suction-controlled wetting–drying cycles to evaluate long-term collapse and swelling behaviour, and validate the proposed strength envelope. Establishing a standardized procedure for suction-controlled triaxial testing and expanding accredited laboratory capacity would also support broader adoption of unsaturated soil mechanics in geotechnical practice.

Author contributions: The authors contributed equally to this work.

Funding: No financial support for the publication of this article.

Acknowledgments: The authors thank WSP Geotechnical Laboratory in Peru, Lima, Peru for providing the facilities for the execution of the tests.

Conflicts of interest: There is no conflict of interest.

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